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# Systematic Evaluation Approach to Support Effective Web Interaction Among Visually Impaired Users

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**Abstract**—Today, Internet has become one of the most useful tools for visually impaired community. They use Internet to help with their daily activities such as communication, information retrieval, online shopping, transportation, and many more using interactive websites. However, with the complexity of current website design, it somehow gives them a hard time. Therefore, it is critical to evaluate the interaction experience perceived by the visually impaired users when interacting with website. Based on analysis of previous research, the evaluation can be measured through three major aspects which are user experience, web accessibility and technology acceptance. Those three aspects are expected to influence the overall judgement of web application itself. This paper is meant to provide some insight on the methodological approach for evaluation of interactive website which emphasis on web accessibility need by the visually impaired users.

**Keywords**—Interaction experience, Web accessibility, Technology acceptance, Evaluation approach

## I. INTRODUCTION

Nowadays, Internet has become necessity. The improvement of the Internet has changed our lifestyle. It has changed the way a website works previously. The website design had evolved tremendously from a static one-way interaction to become more complex which requires more interaction. While some are happy with this changed, there are special need population that having bad experience with it when the websites turn into inaccessible to them. Generally, there are four main type of special needs which are visual, hearing motor and cognitive impairment. Each of this impairment has different requirement with regards to the website design. However, visual impairment and blind (VIB) is the most critical as they are purely relying on the voice whereas website conveys its information through text and display. Many of the usability research has indicated that the VIB are having difficulties to complete all the task assign to them navigating a website [1][2][3]. Thus, there must be something that need be done to help VIB to get better interaction experience. Hence, it comes the importance of the evaluation criteria to measure the overall interaction experience of the VIB users.

UX is of the notion that is used to measure the interaction experience. It is subjective and depending on the person who experiences, which is different from person to person. UX divided the interaction into instrumental and non-instrumental qualities. Both qualities are expected to influence the emotional reaction and the outcome of the interaction process[4]. It also mentioned that situation or context also

influence the interaction outcome. The goodness of a website should not only focusing on the functionalities and completeness, but it should look into overall experience which inclusive the psychological and emotional aspects [5]. In order to deliver a successful website, understanding and evaluating UX is a crucial process [6]. UX also claims that the success of a website was positively influenced by the extent of the high-quality experience promoted to the users.

Web accessibility is crucial aspects which interactive website must adhered to support VIB needs. Accessible website not only give benefit to the disabled users but also normal users. The unified definition of web accessibility is hard to find [7]. However it can be summarized that website should be design to be accessible to all users regardless their disability, so that they can use and contribute to it [8].

The last aspect that should be measured is the acceptance of the technology itself. Successful of technology is important to fully take part in current quick changing modern environment. The most popular and acknowledged approach for studying technology acceptance is using Technology Acceptance Model (TAM). It was pioneered by Davis in last quarter century [9]. TAM measures the behavioral intention to use a website which then will develop the usage behavior based on two main elements which are perceived usefulness and perceived ease of use.

After analyzing the concept of user experience, web accessibility and technology acceptance, it is expected covers the attributes of effective web interaction experience among VIB. Although the analysis was conducted only based on literature, it provides insight on how those three aspects conceptualized and can be integrated together. A conceptual model which integrated the three concept has been proposed in [10]. Details of the model were depicted in figure 1 below. The model's aim to evaluate users' experience and acceptance of news website among the visually impaired users. The selection of the variables are based on the past literature in the respective domain.

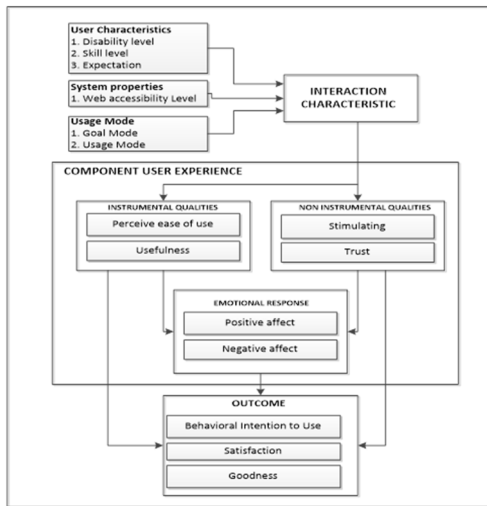


Fig. 1. Conceptual Model of UX, Web Accessibility and TAM [10]

## II. BACKGROUND STUDY OF EACH ASPECTS

The concept of user experience, web accessibility and product acceptance have gained importance. Those three aspects are considered as an experiential component perceived by users inclusive the VIB users. It was expected to have impact on the overall product judgement and acceptance. As we are aware that success and failure of a product is positively influenced by a high-quality experience perceived by their users [6].

### A. User Experience

User experience usually considered as fuzzy and hard to define concept and it was far from having a coherent understanding among previous scholar [11]. However, nearly all of the researchers agree that UX are subjective, context-dependent, and dynamic [12][13] [9].

Subjective referring to the notion that user experience is depending on the person who experiences, which is different from person to person. While context-dependent indicate that user experience is vulnerable to the change of the setting during the interaction happened. The quality on the user experience is affected by the contextual change. Lastly, user experience is dynamic where it is an ongoing phenomenon that changes overtime. User experience is momentary nature that occur before, during and after interaction use which might have a different result. Thus, ISO 9241-210, defined UX as user's perceptions and responses that result from the use and/or anticipated use of a system, product, or service. Users' perceptions and responses include the users' emotions, beliefs, preferences, perceptions, comfort, behaviors, and accomplishments that occur before, during and after use.

There is also a notion claimed that UX is an extended of usability concept [14]. In this notion, UX is classified into three main types [15][12][5]: (a) extension of usability concept. UX can be viewed as an elaboration of satisfaction component of usability by adding effect of emotions, (b) UX primary focus on subjective like emotional measure which did not cover by usability concept and (c) holistic view of user experience where it covers broader interaction result inclusively. It covers the attributes of cognition, attitudes, belief, behavioral intention and affect. With regard of various definition of UX concept it was claimed, the definition of UX

were affected by the researcher's background and interest and could not be used as a common definition [5].

There are various model that represent UX approach and the most commonly used are the CUE Model [16] and UX Model [17]. Both models share the same foundations, as both describe UX based on instrumental and non-instrumental quality perceptions, and also include the emotional reactions of users and the outcome or overall judgement of the artefact.

UX spectrum is broad, as it involved end-to-end interaction process. It can be measured before, during and after interaction occurs. Thus, it requires different approach of evaluations and measurement for different stages of the interaction. The absence of a theoretical framework that links together various UX dimension resulted that there is no specific standard for UX evaluation to be followed [5].

Many research methodologies are available to study UX. Generally, there are two stances on how UX should be studies: qualitative or quantitative[18]. The most frequently used method includes the observation, field studies, surveys, usability studies, interviews, focus group and controlled experiment. However, expert-based method are considered no suitable because UX is very subjective to individuals [15].

Considering that UX is an affective outcome of interaction process, the concept measures the emotional and psychological aspect which user perceived during the interaction. Questionnaire is one of the most popular tools used in most of UX research. This is because of the quality of the data that will be produced with minimum cost. Questionnaire is used as a structure for tester to rate their interaction experience. The questionnaire usually crafted with psychometric scales which measures the subjective nature of UX. Numbers of questionnaires have been developed which capture difference aspect of UX dimension. The most acknowledge questionnaire that frequently used in UX study are AttrakDiff, UEQ, and mCUE [19]. Lastly, the selection of the evaluation method and tools for measuring UX is rely on the researcher objective.

### B. Web Accessibility

The improvement of technology has transformed website from a simple text interface to dynamic and more interactive design. While this move has provided users with more creative and flexible experience, there are special groups of users having hard time with this improvement. Their requirements were being neglected. The website being developed and published without considering the web accessibility aspect.

Basically, there are four main categories of impairment. They are usually associated to visual, hearing, motor, and cognitive impairment. However, the most impacted with inaccessible website is the VIB as most of the website heavily depending on visual cues for its interaction. Typical problem faced among the visually impaired includes difficulties in reading text. This happened when the text and the background were in low contrast, missing alternative text for non-text information and problem navigating website without mouse.

Web accessibility has longed be the attention of World Wide Web Consortium (W3C). W3C crate a special program under its Web Accessibility Initiative (WAI) to deliver a web accessibility standard that can be used by the web developer as a guideline to develop accessible website. The WAI deliver the international standard describing guidelines that involve in accessible website development. Web Content Accessibility

Guidelines (WCAG) is has become widely known standard [20]. However, there are other guideline issued by other organization and companies such as the US Government Section 508 (2018), IBMWeb Accessibility Checklist (IBM2017) ortheBBC (2018). Many of the guideline was inspired by cases highlighted by W3C.

There is various definition of web accessibility found in previous research. Web accessibility can be defined from three perspective; (i) from a user experience perspective, “Web accessibility means that websites, tools, and technologies are designed and developed so that people with disabilities can use them” (ii) from a technical perspective, “Web accessibility means understanding the diverse user needs and translating them into specific technical requirements for designers and developers of websites, tools, and technologies”; (iii) from a production perspective, Web accessibility means integrating accessibility requirements into the design, development, and maintenance processes of websites, tools, and technologies to ensure equal access for people with disabilities, while, at the same time, improving usability and promoting inclusion for all users [21].

Like user experience, web accessibility concept facing the same issue in term of its definition. However, based on analyzing various of technical sources and managed to derived a unified definition of web accessibility as “all people, particularly disabled and older people, can use websites in a range of contexts of use, including mainstream and assistive technologies; to achieve this, websites need to be designed and developed to support usability across these contexts” [8].

Web accessibility evaluation is an assessment of how good an interactive website can be utilized by the disabled users. This is an essential process that for website developer to verify either their website adopting accessibility standard or not. For WCAG standard, it promotes four accessibility principles which are:

TABLE I. WCAG DESIGN PRINCIPLE

| WCAG principles | Description                                                                                                                                                                                                                                                                                           |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Perceivable     | Information and user interface components must be presentable to users in ways they can perceive. This means that users must be able to perceive the information being presented (it can't be invisible to all their senses)                                                                          |
| Operable        | User interface components and navigation must be operable. This means that users must be able to operate the interface (the interface cannot require interaction that a user cannot perform).                                                                                                         |
| Understandable  | Information and the operation of user interface must be understandable. This means that users must be able to understand the information as well as the operation of the user interface (the content or operation cannot be beyond their understanding).                                              |
| Robust          | Content must be robust enough that it can be interpreted reliably by a wide variety of user agents, including assistive technologies. This means that users must be able to access the content as technologies advance (as technologies and user agents evolve, the content should remain accessible) |

The evaluation of web accessibility can be performed using automated testing tools. This method allows the application to scan through the website pages and verify the compliance for specific set of guidelines. For example, if the tools are going to check against WCAG standard, the result will be based on the four principles as per stated in table above. This approach usually used as first accessibility test due to the process fast, cost effective and reliable to the extent of producing reproducible evaluation results [22]. The automatic evaluation tools report their result in metrics. It will then produce a scoring based on the failure and success detected. Some of the tools provide general recommendation on how to improve the accessibility the website. The effectiveness of a tools depends on the representation given to the guideline. It was suggested by [20] to use three different automated tools as the result might be vary. This is due to the code and specification that use in different automated tools. There are many tools that are currently available in the market. It was reported by [23], Accessibility Valet, AChecker, Cynthia Says, EvalAccess, FAE, MAGENTA, OCAWA, TAW, WAVE and Web Accessibility Checker are the top-10 free tools. A comprehensive list of available web accessibility evaluation tools can be found on W3C WAI website. Beside automated tools, some studies which involve the investigation of accessibility barriers tend to use qualitative such as user testing [24]. However, this method seldom be used due to the time consuming and need more financial aids.

Despite a growing awareness of web accessibility, people are still experiencing barriers to access the interactive artifact. Emphasizing web accessibility design approach will benefit everyone not only people with disabilities. This is what is referred to as “accessible Web design,” “design for all,” or “universal design.” Accessible interactive product will not only beneficial for the impairment users but also for the normal users.

### C. Technology Acceptance

With the growing and advancement of technology, it has become necessity to understand the reasons behind the technology acceptance nor rejection. Generally, acceptance is defined as “an antagonism to the term refusal and means the positive decision to use an innovation” [25]. Based on the number citation, Technology Acceptance Model (TAM) has been proven as the most widely used model to evaluate technology acceptance especially website [26] [9]. TAM was first introduced more than a century ago that expected to predict the technology acceptance based on the perceive usefulness and perceived ease of use of an interactive website [27]. The model kept on developing through the years of research and experience various extension. The extension implied various additional external factor which expected influence the user’ intention to use and then will develop the usage behavior.

TAM was grounded from theories of reasoned action and planned behavior which is originally from psychology field. In TAM, a person’s perception of the usefulness and ease of use of a system determines the person’s attitude toward using the system, which in turn determines whether the person forms an intention to use the system and develop a usage behavior to use it again. Perceived usefulness relates to how much a person trust that using a system will improve their job performance. While perceived ease of use is related to the extent of user’s trust when using the system is straight forward

and effortless. With combination of both aspects, it is expected to give result on the acceptance of the interactive artifact.

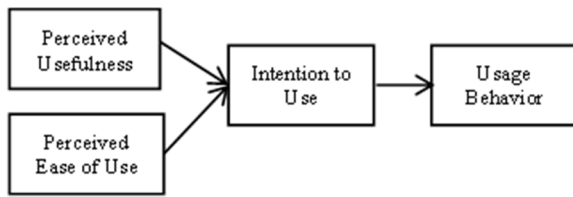


Fig. 2. Technology Acceptance Model [8]

The method to evaluate technology acceptance will be not much different with UX evaluation. This is due to the nature of it where it is considered as very individual. Expert based method is not suitable as it does not picture the actual users of the interactive product which will develop inaccurate result.

Referring to multiple research on TAM, questionnaire is the most used tools [28][29][30]. Questionnaire that measures the construct being distributed to the participant after user the product. TAM must be measured after a person using a product. It can be measure right after first use, or it can also be measured after multiple use. The evaluation method will be much depending on the objective of the research itself.

After analyzing the concept of user experience, web accessibility and technology acceptance, it is expected covers the attributes of effective web interaction experience among VIB. Although the analysis was conducted only based on literature, it provides insight on how those three aspects conceptualized and can be integrated together. A conceptual model which integrated the three concept has been proposed in [10]. The model's aim to evaluate users' experience and acceptance of news website among the visually impaired users. The selection of the variables were based on the past literature in the respective domain

### III. THE EVALUATION: OUR APPROACH

Undergoing the analysis on the past research and performed comparative analysis of the three concepts, we believed that user experience, web accessibility and technology accepted cover the attributes of effective VIBs' web interaction experience. Thus, the combination on the three concepts expected to give insight to website designer to promote more accessible website.

In order to proceed future, user study should be the suitable method to evaluate those concepts. User research focus on the process of understanding the impact of website design toward the users. It focuses on understanding users' behavior, need and motivations. However, there are no tools that has established for aggregating the assessment of the three concepts

Therefore, we are adapting the meCUE 2.0 questionnaire as a tool for the assessment. meCUE 2.0 is one of the establish instrument that provide a comprehensive structure to access those three concepts. It was based on the theory of Comprehensive User Experience (CUE) mode. It has 34 items which covers the component of the model and its sub-construct.[31]. It separated the scores into five different modules.

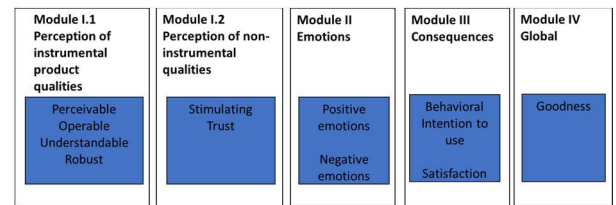


Fig. 3. meCUE 2.0 questionnaire modules

meCUE 2.0 consist of five separately validated modules. It distinguishes between the perception of instrumental and non-instrumental qualities. Instrumental qualities represent the functionality aspect while the non-instrument qualities are referring to the psychological aspect of the interaction. Emotion's module captures the user emotions which consist of negative and positive emotions. Module III represent the consequences where behavioral intention to use and satisfaction will be measured. Lastly module IV represents the global where it represents the overall evaluation of the interactive product. The above Figure 2 depict the summarize of each construct for every module in meCUE 2.0 questionnaire

To adapt meCUE 2.0 questionnaire into research objective, selection of construct for each module has been done carefully from user experience, web accessibility and technology acceptance domain. The mixture of the construct from those three attributes are expected to be able to measure the effective web interaction experience among VIB users. The Table II below describe the construct's definition in detail.

TABLE II. EVALUATION CONSTRUCTS AND DEFINITION

| Instrumental Qualities     |                 |                                                                                                                                                               |
|----------------------------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.                         | Perceivable     | Information available on the web should be presentable to all users.                                                                                          |
| 2.                         | Operable        | User interface features is operable by the users.                                                                                                             |
| 3.                         | Understandable  | User must be able to understand and predict how the user interface of the website operate.                                                                    |
| 4.                         | Robust          | The website must be able to be used robustly.                                                                                                                 |
| Non-Instrumental Qualities |                 |                                                                                                                                                               |
| 5.                         | Stimulating     | Feeling of excitement or causing great emotional feeling towards the website. Most visually impaired users perceived web accessibility as stimulating [7]     |
| 6.                         | Trust           | Related to the trustworthy and comfort feeling towards the websites. According to [32], there are significant correlation between trust and web accessibility |
| Emotional Response         |                 |                                                                                                                                                               |
| 7.                         | Positive affect |                                                                                                                                                               |

|         |                             |                                                                                                                                                                  |
|---------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8.      | Negative affect             | Subjective feeling accompanied by specific physiological reaction and express behavior. Based on [16] emotional response may affect the appraisal of the system. |
| Outcome |                             |                                                                                                                                                                  |
| 9.      | Behavioral Intention to Use | Related to the user's willing to use and accept the system [27].                                                                                                 |
| 10.     | Satisfaction                | Related to feeling of being pleased about the confirmation of the prospects of a desirable event [17].                                                           |
| 11.     | Goodness                    | The overall interaction quality [33] whether it is good or bad. For this research it is more related to the accessibility of the web.                            |

Based on the construct definition, for instrumental qualities we are adopting the questions from WCAG design principles to evaluate the construct. The combination on mECUE 2.0 questionnaire with item from WCAG is expected should be able to evaluate interactive website based on user experience, web accessibility and technology acceptance.

#### IV. CONCLUSION AND FUTURE WORK

This paper presented the detail background study on three aspect which are user experience, web accessibility and technology acceptance. There three aspect is important to be evaluated on interactive website to deliver more accessible websites. An instrument has been proposed which covers all three aspects.

Next, we are planning to perform control experiment with VIB user navigating websites with different level of web accessibility. The proposed questionnaire will be distributed to the responded after they finishes the navigation task. Statistical data analysis will be conducted to find the relationship between each construct. The structured equation modelling method expected to be used to access the cause-effect, path connection within the construct.

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